

Strictly Confidential

Web SDK

October 08, 2021



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1. BACKGROUND

This note describes the mode/ manner of the technical integration between the Merchant and BillDesk platform to enable one-time payment & recurring payment workflows via payment methods such as Cards/ UPI/ eNACH.

2. TYPES OF SDK

Following are the Web SDK's that are available for the multiple use-cases:

- (a) Payment SDK
- (b) Payment + Mandate Setup SDK
- (c) Only Mandate Setup SDK
- (d) Modify/Delete Mandate SDK

3. PREREQUISITES TO GET STARTED

- Obtain your Merchant ID (MID) from BillDesk
- Merchant's Server public IP address to be whitelisted at BillDesk
- **Create Order API** has to be called using JWS-HMAC for Payment SDK and Payment + Mandate Setup SDK
- **Create Mandate Token API** to be called using JWS-HMAC for Only Mandate Setup SDK
- **Update Mandate Token API** to be called using JWS-HMAC for Modify/Delete Mandate Setup SDK

4. HEADERS and AUTHENTICATION

JWS represents signed content using JSON data structures and base64url encoding. The representation consists of three parts: the JWS Header, the JWS Payload, and the JWS Signature. The three parts are base64url-encoded for transmission, and typically represented as the concatenation of the encoded strings in that order, with the three strings being separated by period ('.') characters.

The JWS Header describes the signature method and parameters employed. The JWS Payload is the message content to be secured. The JWS Signature ensures the integrity of both the JWS Header and the JWS Payload.

JWS-HMAC is a framework intended to provide a method to securely transfer claims (such as authorization information) between parties.

- Peer-to-Peer communication will happen over HTTPS (TLS1.2).
- JWS-HMAC – In this implementation a secret key/HMAC key will be shared by BillDesk with the merchant, using which the merchant will sign the request and then when request comes to BillDesk, it'll verify the request using the same secret key. The same key is used at both ends.
- Content-Type: application/jose (content-type remains jose only) Body uses compact serialization format as per JWS (i.e., RFC 7515) and the payload is JSON.
- The JWS header fields to be included are as follows:
 - The algorithm ("alg" field) with value "HS256".
 - Custom header "clientid" is the clientid value provided by BillDesk

Request Headers

With each request, following headers are required:

Attribute	Type	Description
Content-Type	String	Request Content-Type to take the values application/jose
Accept	String	Accept Response Content-Type to take the values application/jose
BD-Traceid	String	Traceid is used for idempotency and the request with the same Traceid within the day will be rejected. Traceid can be alphanumeric without any spaces or special characters and should be a maximum size of 35
BD-Timestamp	String	epoch timestamp of the server

The process to generate JWS-HMAC request is as below:

Step 1 : Receive a clientid and a secretkey from BillDesk. The clientid identifies the client sending the requests, and the secretkey is used to generate the hash for the request.

Example :

```
{  
  "alg": "HS256",  
  "clientid" : "mercclientid"  
}
```

Step 2 : Create the Payload (JSON) -

```
{  
  "mercid": "BDMERCID",  
  "orderid": "order45608988",  
  "amount": "300.00",  
  "order_date": "2021-03-05T10:59:15+05:30",  
  "currency": "356",  
  "ru" : "https://merchant.com",  
}
```

[illegible]

With each response, you will receive the following response headers:

Attribute	Type	Description
Content-Type	String	Request Content-Type to take the values application/jose
BD-Traceid	String	Same Traceid as passed in request

BD-Timestamp	String	Same timestamp value as passed in request
--------------	--------	---

5. SDK INTEGRATION

Include “billdesksdk.esm.js”, “billdesksdk.js” & “billdesksdk.css” as part of your checkout page.

Sandbox URL: <https://uat.billdesk.com/jssdk/v1/dist/>

Production URL: <https://pay.billdesk.com/jssdk/v1/dist/>

Sample Code:

```
<script type="module"
src="https://uat.billdesk.com/jssdk/v1/dist/billdesksdk/billdesksdk.esm.js"></script>
<script nomodule="" src="https://uat.billdesk.com/jssdk/v1/dist/billdesksdk.js"
"></script>
<link href="https://uat.billdesk.com/jssdk/v1/dist/billdesksdk/billdesksdk.css"
rel="stylesheet">
```

5.1. Payment SDK / Payment + Mandate Setup SDK

You will need to pass below arguments when invoking the SDK to launch the checkout experience for the customer.

Argument	Description
<i>logo</i>	base64 image of merchant's logo
<i>flowType</i>	Set value as “payments”
<i>childWindow</i>	Set value as “true”
<i>retryCount</i> (optional)	Number of retry attempts you want the customer to be able to get. Say the earlier attempt to complete the transaction failed, the customer will get a retry option in the SDK user interface.
<i>returnUrl</i> (optional)	Merchant return URL ((If this parameter is passed, the merchant will receive a redirection on this URL on payment completion, otherwise the merchant page will receive a function callback for childwindow))
<i>prefs</i>	Overrides payment categories of the preferences in given order
<i>responseHandler</i>	JavaScript call back function to receive the transaction's metadata after completion of the transaction journey
<i>bdOrderId</i>	Order id generated using Create Order API
<i>merchantId</i>	Merchant id received from BillDesk once you're on boarded

authToken	Token generated using Create Order API
------------------	--

5.1.1. Create Order API

This API is applicable when using the Payment SDK / Payment + Mandate Setup SDK

- Merchant must initiate the Create Order API using JWS-HMAC
- The arguments **bdOrderId** & **authToken** is returned as part of the API response from BillDesk

Attributes	Mandatory/ Optional/ Conditional	Description
orderid	Mandatory	Unique orderid generated by the merchant to identify the transaction
mercid	Mandatory	Unique identifier as defined by BillDesk for each merchant
order_date	Mandatory	Merchant order generation date and time in YYYY-MM-DDThh:mm:ssTZD format
amount	Mandatory	Order amount in two decimals
currency	Mandatory	ISO currency of the transaction amount
ru	Mandatory	Merchant return url**
additional_info	Optional	Array of 10 additional_info values that can be attached to the transaction
itemcode	Mandatory	Itemcode value as provided by BillDesk, with a default value DIRECT
recurrence_rule	Conditional	Applicable for UPI. Values can be on/ before/ after
debit_day	Conditional	• Applicable for UPI. Value that is specified depending upon the frequency. The values are as follows: ○ For a Weekly frequency the value for this field must be between 1 to 7 where 1 means Monday, 2 is Tuesday etc. ○ Fortnightly (starting from 1 to 14/15/16 [depending on the month]) ○ Monthly/ Bi monthly/ Quarterly/ Half-yearly/ Yearly [1 – 30/31 (if 30/31 is not available, the last day of the month will be considered for the debit)]
mandate_required	Conditional	For Payment + Mandate flow, if on same mercid requires optional as well as compulsory flow, then need to pass this flag as Y – for compulsory and N – for optional flow; provided prior set-up is done on the mercid

settlement_lob	Optional	Settlement line of business preconfigured by BillDesk for funds settlement to merchant account
mandate (object)	Conditional	Mandate object – to be passed for Payment + Mandate flow
customer (object)	Optional	Customer object
device (object)	Mandatory	Device object
invoice (object)	Optional	Applicable for UPI QR (to include GST details) • invoice_number • invoice_display_number • customer_name • invoice_date • gst_details ○ cgst ○ sgst ○ igst ○ gst ○ cess ○ gstincentive ○ gstpct

**** Note:** If *childWindow* flag will be set as **true** while launching SDK then RU should have only onload close event.

Sample Code below:

```
<html><head><title>Billdesk</title></head><body onload="window.close();"></body></html>
```

Response

Returns the order object.

Sandbox URL: https://pguat.billdesk.io/payments/ve1_2/orders/create

(A) SAMPLE (FOR PAYMENT ONLY SDK)

Headers:

```
content-type: application/jose
bd-timestamp: 20200817132207
accept: application/jose
bd-traceid: 20200817132207ABD1K
```

Request: (encrypted & signed using JWS-HMAC)

```
{
  "mercid": "BDMERCID",
  "orderid": "TSSGF43214F",
  "amount": "300.00",
  "order_date": "2020-08-17T15:19:00+0530",
  "currency": "356",
  "ru": "https://www.example.com/merchant/api/pgresponse",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2"
  },
  "itemcode": "DIRECT",
  "invoice": {
    "invoice_number": "MEINVU11111221133",
    "invoice_display_number": "11221133",
    "customer_name": "Tejas",
    "invoice_date": "2021-09-03T13:21:5+05:30",
    "gst_details": {
      "cgst": "8.00",
      "sgst": "8.00",
      "igst": "0.00",
      "gst": "16.00",
      "cess": "0.00",
      "gstincentive": "5.00",
      "gstpct": "16.00",
      "gstin": "12344567"
    }
  },
  "device": {
    "init_channel": "internet",
    "ip": "202.149.208.92",
    "mac": "11-AC-58-21-1B-AA",
    "imei": "990000112233445",
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0",
    "accept_header": "text/html",
    "fingerprintid": "61b12c18b5d0cf901be34a23ca64bb19"
  }
}
```

Response: (Encrypted & Signed using JWS-HMAC will need decryption & sign verification)

```
{
  "objectid": "order",
  "mercid": "BDMERCID",
  "amount": "300.00",
  "ru": "https://www.demourlmerchant.com/response",
  "orderid": "6yn4uhuz9e9bcw9qo6r6v4",
  "order_date": "2020-08-17T15:19:00+0530",
  "bdorderid": "O AFC19XTFD8TSP",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2"
  },
}
```

```
"next_step":"redirect",
"itemcode":"DIRECT",
"createdon":"2021-03-11T19:23:24+05:30",
"currency":"356",
"links":[
{
"method":"GET",
"rel":"self",
"href":"https://www.sampleurl.com/request"
},
{
"headers":{
"authorization":"OToken
DEDC1071B77800A146B6E8D2530E0429E76520C151B40CC3325D8B6D9242CBA3A6BFA64
3E7E5596FBEBAE0F46A1FB1BCD099EBC1F59DCD82F390B6BC45FCE036F37F7F589BD687
A691E1378F1FF432331C62E7E641E857C8F8A405A4BFE2F01B1EB8F3C69817D45F5DDE9
DEE346ACABA1B7208DECA9E43CCE7AB3761553E23D9CB36A870C1819C15C7C4B1CFE280
2DFD05F651AA537AB81787.4145535F55415431"
},
"valid_date":"2020-08-17T15:49:00+0530",
"method":"POST",
"rel":"redirect",
"href":"https://www.billdesk.com/pgi/MerchantPayment/",
"parameters":{
"mercid":"BDMERCID",
"bdorderid":"OAFCl9XTFD8TSP"
}
},
"status":"ACTIVE",
"invoice":{
"invoice_number":"MEINVU111111221133",
"invoice_display_number":"11221133",
"customer_name":"Tejas",
"invoice_date":"2021-09-03T13:21:5+05:30",
"gst_details":{
"cgst":"8.00",
"sgst":"8.00",
"igst":"0.00",
"gst":"16.00",
"cess":"0.00",
"gstincentive":"5.00",
"gstpct":"16.00",
"gstin":"12344567"
}
},
"device":{
"accept_header":"text/html",
"ip":"202.149.208.92",
"fingerprintid":"61b12c18b5d0cf901be34a23ca64bb19",
"imei":"990000112233445",
"init_channel":"internet",
"mac":"11-AC-58-21-1B-AA",
"user_agent":"Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0)
Gecko/20100101 Firefox/51.0"
}
```

```
}
```

(B) SAMPLE (FOR PAYMENT + MANDATE SDK)

Headers:

```
content-type: application/jose  
bd-timestamp: 20200817132207  
accept: application/jose  
bd-traceid: 20200817132207ABD1K
```

Request: (encrypted & signed using JWS-HMAC)

```
{  
  "mercid": "BDMERCID",  
  "orderid": "order919217288",  
  "amount": "2.00",  
  "order_date": "2021-09-03T09:59:06+05:30",  
  "currency": "356",  
  "itemcode": "DIRECT",  
  "ru": "https://merchant.com",  
  "additional_info": {  
    "additional_info1": "Details1",  
    "additional_info2": "Details2"  
  },  
  "device": {  
    "init_channel": "internet",  
    "ip": "17.233.107.92",  
    "mac": "11-AC-58-21-1B-AA",  
    "imei": "990000112233445",  
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0",  
    "accept_header": "text/html",  
    "fingerprintid": "61b12c18b5d0cf901be34a23ca64bb19"  
  },  
  "mandate": {  
    "mercid": "BDMERCID",  
    "currency": "356",  
    "amount": "200.00",  
    "customer_refid": "cust9199028201",  
    "subscription_refid": "Sub92882992782701",  
    "subscription_desc": "Term insurance",  
    "start_date": "2021-09-30",  
    "end_date": "2021-12-04",  
    "frequency": "year",  
    "amount_type": "max",  
    "recurrence_rule": "on",  
    "debit_day": "6"  
  },  
  "invoice": {  
    "invoice_number": "MEINVU11111221133",  
  }  
}
```

```
"invoice_display_number":"11221133",
"customer_name":"Tejas",
"invoice_date":"2021-09-03T13:21:5+05:30",
"gst_details":{
  "cgst":"8.00",
  "sgst":"8.00",
  "igst":"0.00",
  "gst":"16.00",
  "cess":"0.00",
  "gstincentive":"5.00",
  "gstpct":"16.00",
  "gstin":"12344567"
}
}
```

Response: (Encrypted & Signed using JWS-HMAC will need decryption & sign verification)

```
{
  "objectid":"order",
  "orderid":"order919217288",
  "bdorderid":"OAJZ19XTITYVUV",
  "mercid":"BDMERCID",
  "order_date":"2021-09-03T09:59:06+05:30",
  "amount":"2.00",
  "currency":"356",
  "ru":"https://merchant.com",
  "additional_info":{
    "additional_info1":"Details1",
    "additional_info2":"Details2"
  },
  "itemcode":"DIRECT",
  "createdon":"2021-09-03T18:54:12+05:30",
  "next_step":"redirect",
  "links":[
    {
      "href":"https://www.billdesk.com/pgi/v1_2/orders/order919217288",
      "rel":"self",
      "method":"GET"
    },
    {
      "href":"https://pguatweb.billdesk.io/pgtxnsimulator/v1_2/transactions/orderform",
      "rel":"redirect",
      "method":"POST",
      "parameters":{
        "mercid":"BDMERCID",
        "bdorderid":"OAJZ19XTITYVUV"
      },
      "valid_date":"2021-09-03T19:24:12+05:30",
      "headers":{
        "authorization":"OToken
6C0E4182148225FE963C491087F7EE84464736F25C3FDBBC55BC9C6586B9AA00421998A
7F134D223CDD547612F51DE7DA7F3EFD57A8D441872BF4B89EDB0AB371DD9410D8923E7
C5DF50111E854108989CA149480ABC978D03AA9C8FC84C934286C7488F172B8E6C3623E"
```

```

84B7F4DFDFC0C29188AFDA722986C67AFF84AF9389D1D3E7367C19500C846EC936C3C6B
07620E68765946958CB8D9.4145535F55415431"
}
}
],
"mandate":{
"mercid":"BDMERCID",
"customer_refid":"cust9199028201",
"subscription_refid":"Sub92882992782701",
"subscription_desc":"Term insurance",
"amount_type":"max",
"start_date":"2021-09-30",
"end_date":"2021-12-04",
"frequency":"year",
"currency":"356",
"amount":"200.00",
"status":"initiated",
"mandate_tokenid":"NA1DF800065707",
"recurrence_rule":"on",
"debit_day":"6",
"createdon":"2021-09-03T18:54:12+05:30"
},
"status":"ACTIVE",
"invoice":{
"invoice_number":"11221133",
"invoice_display_number":"11221133",
"invoice_date":"2021-09-03T13:21:5+05:30",
"customer_name":"Tejas",
"gst_details":{
"cgst":"8.00",
"sgst":"8.00",
"igst":"0.00",
"gst":"16.00",
"cess":"0.00",
"gstincentive":"5.00",
"gstpct":"16.00",
"gstin":"12344567"
}
}
}
}

```

5.2. Only Mandate Setup SDK

You will need to pass below arguments when invoking the SDK to launch the checkout experience for the customer.

Argument	Description
<i>logo</i>	base64 image of merchant's logo
<i>flowType</i>	Set value as "emandate"
<i>childWindow</i>	Set value as "true"

<i>retryCount</i> (optional)	Number of retry attempts you want the customer to be able to get. Say the earlier attempt to complete the transaction failed, the customer will get a retry option in the SDK user interface.
<i>returnUrl</i> (optional)	Merchant return URL ((If this parameter is passed, the merchant will receive a redirection on this URL on payment completion, otherwise the merchant page will receive a function callback for childwindow))
<i>prefs</i>	Overrides payment categories of the preferences in given order
<i>responseHandler</i>	JavaScript call back function to receive the transaction's metadata after completion of the transaction journey
<i>mandateTokenId</i>	Order id generated using Create Mandate Token API
<i>merchantId</i>	Merchant id received from BillDesk once you're on boarded
<i>authToken</i>	Token generated using Create Mandate Token API

5.2.1. Create Mandate Token API

This API is applicable when using the Mandate Setup SDK

- Merchant must initiate the Create Mandate Token API using JWS-HMAC
- The arguments **mandateTokenId** & **authToken** is returned as part of the API response from BillDesk

Attributes	Mandatory/ Optional/ Conditional	Description
mercid	Mandatory	Unique identifier as defined by BillDesk for each merchant
customer_refid	Mandatory	Unique reference of the customer in the merchant system, and can be linked to different subscriptions of the same customer
subscription_refid	Mandatory	Unique reference assigned by merchant for the subscription
subscription_desc	Optional	Customer friendly description for the subscription
amount	Conditional	Subscription amount in two decimals (if Merchant is capturing mandate details)
currency	Conditional	ISO currency of the transaction amount
frequency	Conditional	Frequency at which the subscription debits will be processed. (if Merchant is capturing mandate details) Possible values are • week - weekly • mnth - monthly • bimn – bi-monthly • qurt - quarterly • bian – bi-annually

		- year - yearly - adho – adhoc
recurrence_rule	Conditional	Applicable for UPI. Values can be on/ before/ after
debit_day	Conditional	- Applicable for UPI. Value that is specified depending upon the frequency. The values are as follows: - For a Weekly frequency the value for this field must be between 1 to 7 where 1 means Monday, 2 is Tuesday etc. - Fortnightly (starting from 1 to 14/15/16 [depending on the month]) Monthly/ Bi monthly/ Quarterly/ Half-yearly/ Yearly [1 – 30/31 (if 30/31 is not available, the last day of the month will be considered for the debit)]
amount_type	Conditional	Indicates whether subscription amount is fixed or has a maximum limit. The following values are possible. maximum - indicates the maximum amount that can be charged for the subscription, fixed – indicates the fixed amount to be charged for the subscription(if Merchant is capturing mandate details)
start_date	Conditional	Subscription start date in yyyy-mm-dd format(if Merchant is capturing mandate details)
end_date	Conditional	Subscription end date in yyyy-mm-dd format(if Merchant is capturing mandate details)
ru	Mandatory	Merchant return url**
customer (object)	Optional	Customer object
device (object)	Mandatory	Device object

**** Note:** If *childWindow* flag will be set as **true** while launching SDK then RU should have only onload close event.

Sample Code below:

```
<html><head><title>Billdesk</title></head><body onload="window.close();"></body></html>
```

Response

Returns the mandate token object.

Sandbox URL: https://pguat.billdesk.io/pgsi/ve1_2/mandatetokens/create

- **SAMPLE (FOR MANDATE ONLY SDK)**

Headers:

```
content-type: application/jose
bd-timestamp: 20201130102651
accept: application/jose
bd-traceid: 20201130102651SWRHV
```

Request: (encrypted & signed using JWS-HMAC)

```
{
  "mercid": "BDMERCID",
  "customer_refid": "cust980823yr99",
  "subscription_refid": "Sub980uh99",
  "subscription_desc": "9800000111",
  "currency": "356",
  "frequency": "year",
  "amount_type": "max",
  "amount": "2.00",
  "start_date": "2021-09-30",
  "end_date": "2021-12-04",
  "recurrence_rule": "on",
  "debit_day": "6",
  "customer": {
    "first_name": "JohnDoe",
    "last_name": "Doe",
    "mobile": "9800000000",
    "mobile_alt": "9800000000",
    "email": "john.doe@somedomain.com",
    "email_alt": "john.doe@somedomain.com"
  },
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2",
    "additional_info3": "123456789"
  },
  "device": {
    "init_channel": "internet",
    "ip": "17.233.107.92",
    "mac": "11-AC-58-21-1B-AA",
    "imei": "990000112233445",
    "user_agent": "Mozilla/Firefox",
    "accept_header": "text/html",
    "fingerprintid": "61b12c18b5d0cf901be34a23ca64bb19"
  },
  "ru": "https://www.google.com"
}
```

Response: (Encrypted & Signed using JWS-HMAC, will need decryption & sign verification)

```
{
  "objectid": "mandate_token",
  "mercid": "BDMERCID",
  "mandate_tokenid": "NA081900006725",
  "createdon": "2021-03-12T10:26:51+05:30",
  "amount": "200.00",
  "customer_refid": "cust1301026513HOGD",
  "subscription_refid": "Sub130102651FTFGA",
  "subscription_desc": "Family Plan",
  "currency": "356",
  "frequency": "year",
  "amount_type": "max",
  "start_date": "2021-03-12",
  "end_date": "2021-12-04",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2",
    "additional_info3": "Details3",
    "additional_info4": "Details4",
    "additional_info5": "Details5",
    "additional_info6": "Details6",
    "additional_info7": "Details7",
    "additional_info8": "Details8",
    "additional_info9": "Details9",
    "additional_info10": "Details10"
  },
  "next_step": "redirect",
  "links": [
    {
      "headers": null,
      "valid_date": null,
      "method": "POST",
      "rel": "self",
      "href": "https://www.demourl/initiate",
      "parameters": null
    },
    {
      "headers": {
        "authorization": "OToken
E7DD329EB2682F5D7AE968BA92C5CF4FC669B71F713099CD081B9438
84161B5F93D33E7F2E1D561607573FA4568FC34A1D82461DF546BCF8
DD677D6E5F7DB5665632E9EDAD0036013A10E632808E922E52461890
EF99229366F909DD18980371D9926C37F86C21E7208F150B28D309F6
58D5C537FD2983B5508FEB2E0FEA1BFE4CD836581EA707F1228107C8
4D4783B73A0C71A03B6F51906054.4145535F55415431"
      },
      "valid_date": "2020-11-30T10:56:51+05:30",
      "method": "POST",
      "rel": "redirect",
      "href": "https://www.sampleurl.com/sdk/",
      "parameters": {
        "mercid": "BDMERCID",
        "mandate_tokenid": "NA081900006725"
      }
    }
  ]
}
```

```

    }
  },
  ],
  "device":{
    "accept_header":"text/html",
    "init_channel":"internet",
    "ip":"202.149.208.92",
    "mac":"11-AC-58-21-1B-AA",
    "imei":"990000112233445",
    "user_agent":"Mozilla/Firefox",
    "fingerprintid":"61b12c18b5d0cf901be34a23ca64bb19"
  },
  "customer":{
    "email_alt":"john.doe@somedomain.com",
    "customer_refid":"cust1301026513HOGD",
    "mobile_alt":"9800000000",
    "mobile":"9800000000",
    "last_name":"Doe",
    "first_name":"John",
    "email":"john.doe@somedomain.com"
  },
  "status":"initiated"
}

```

5.3. Modify/Delete Mandate SDK

You will need to pass below arguments when invoking the SDK to launch the checkout experience for the customer.

Argument	Description
<i>logo</i>	base64 image of merchant's logo
<i>flowType</i>	Set value as "modify_mandate"
<i>childWindow</i>	Set value as "true"
<i>retryCount</i> (optional)	Number of retry attempts you want the customer to be able to get. Say the earlier attempt to complete the transaction failed, the customer will get a retry option in the SDK user interface.
<i>returnUrl</i> (optional)	Merchant return URL ((If this parameter is passed, the merchant will receive a redirection on this URL on payment completion, otherwise the merchant page will receive a function callback for childwindow))
<i>responseHandler</i>	JavaScript call back function to receive the transaction's metadata after completion of the transaction journey
<i>mandateTokenId</i>	Order id generated using Update Mandate Token API
<i>merchantId</i>	Merchant id received from BillDesk once you're on boarded
<i>authToken</i>	Token generated using Update Mandate Token API

5.3.1. Update Mandate Token API

This API is applicable when using the Modify/Delete Mandate Setup SDK

- Merchant must initiate the Update Mandate Token API using JWS-HMAC
- The arguments **mandateTokenId** & **authToken** is returned as part of the API response from BillDesk

Attributes	Mandatory/ Optional/ Conditional	Description
mercid	Mandatory	Unique identifier as defined by BillDesk for each merchant
mandateid	Mandatory	Unique mandateid generated by BillDesk for the associated mandate
action	Optional	If merchant wants specific action. Possible values : modify delete
mandate_edit	Optional	Object – if merchant captures the fields to be edited at their end they can pass (amount, frequency, amount_type, end_date)
ru	Mandatory	Merchant return url**
customer (object)	Optional	Customer object
device (object)	Mandatory	Device object

**** Note:** If **childWindow** flag will be set as **true** while launching SDK then RU should have only onload close event.

Sample Code below:

```
<html><head><title>Billdesk</title></head><body onload="window.close();"></body></html>
```

Response

Returns the mandate token object.

Sandbox URL: https://pquat.billdesk.io/pgsi/v1_2/mandatetokens/update/create

■ SAMPLE REQUEST AND RESPONSE

Headers:

```
content-type: application/jose
bd-timestamp: 20201130102651
accept: application/jose
bd-traceid: 20201130102651SWRHV
```

Request: (encrypted & signed using JWS-HMAC)

```
{
  "mercid": "BDMERCID",
  "mandateid": "MA0ED200052103",
  "action": "modify",
  "mandate_edit": {
    "amount": "1000.00"
  },
  "ru": "http://localhost:3000/api/pgresponse/modal/redirect"
}
```

Response: (Encrypted & Signed using JWS-HMAC, will need decryption & sign verification)

```
{
  "objectid": "mandate_token",
  "mandate_tokenid": "NA0D0800060183",
  "mandateid": "MA0ED200052103",
  "mercid": "BDMERCID",
  "customer_refid": "cus890987067792335",
  "subscription_refid": "12345678901234535",
  "subscription_desc": "Hola",
  "start_date": "2021-06-09",
  "end_date": "2021-12-23",
  "amount": "200.10",
  "amount_type": "maximum",
  "currency": "356",
  "frequency": "adho",
  "createdon": "2021-08-20T13:07:02+05:30",
  "ru": "http://localhost:3000/api/pgresponse/modal/redirect",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2"
  },
  "customer": {
    "customer_refid": "cus890987067792335"
  },
  "device": {
    "fingerprintid": "61b12c18b5d0cf901be34a23ca64bb19",
    "init_channel": "internet",
    "ip": "17.233.107.92",
    "mac": "11-AC-58-21-1B-AA",
    "imei": "990000112233445",
    "user_agent": "Mozilla/Firefox",
    "accept_header": "text/html"
  }
}
```

```

},
"status": "initiated",
"action": "modify",
"payment_method_type": "card",
"mandate_edit": {
  "amount": "1000.00"
},
"next_step": "redirect",
"links": [
  {
    "href": "https://pguat.billdesk.io/pgsi/v1_2/mandates/token/initiate",
    "rel": "self",
    "method": "POST",
    "parameters": null,
    "valid_date": null,
    "headers": null
  },
  {
    "href": "https://www.sampleurl.com/sdk/",
    "rel": "redirect",
    "method": "POST",
    "parameters": {
      "mercid": "BDMERCID",
      "mandate_tokenid": "NA0D0800060183"
    },
    "valid_date": "2021-08-20T13:37:02+05:30",
    "headers": {
      "authorization": "OToken
18183EEED380A39203B26A9230F41F0B20943A2F3045318060C2ED64
9E63BCA26F29BB9E9119318F6F3E7204DE7849F138648AB376B0F544
40DFA3F4D876F51C3E42E5B79B575393B540F86987E7A5BBD4241ADF
9F83A19E90E37E296BD92EC0588565D1C15FE599E4178E94C097BBF2
5FA2B85B3ED89BB5CD0966FCEF320A8DB82B80BCF5241A5C50382A79
FFBD6338C5D46C4836853E0A26.4145535F55415431"
    }
  }
]
}

```

6. Launching the SDK

To launch the SDK you need to invoke the below JavaScript function passing the required arguments as config object shown below.

6.1. Payment SDK / Payment + Mandate Setup SDK

Prepare the flow config object

```

var flow_config = {
  merchantId: "BDMERCID",

```

```
bdOrderId: "TSFFDHWE",
authToken: "OToken FDD5C104249A4CDAE734623",
childWindow: true,
returnUrl: "http://localhost/api/pgresponse/modal/redirect",
retryCount: 3,
prefs: {"payment_categories": ["card", "emi"]}
}
```

Parameter	Description	Classification
merchantId	Merchant id provided by BillDesk	mandatory
bdOrderId	As received in <i>Create Order API</i> response	mandatory
authToken	As received in <i>Create Order API</i> response	mandatory
childWindow	true	mandatory
returnUrl	Merchant return url ((If this parameter is passed, the merchant will receive a redirection on this URL on payment completion, otherwise the merchant page will receive a function callback for childwindow))	optional
retryCount	count for retry (default count: 3)	optional
prefs	payment categories list which need to be displayed in specific order ["card", "emi", "nb", "upi", "wallets", "qr", "gpay"]	optional

6.2. Only Mandate Setup SDK

Prepare the mandate flow config object

```
var mandate_flow_config = {
  merchantId: "BDMERCID",
  mandateTokenId: "TSFFDHWE",
  authToken: "OToken FDD5C104249A4CDAE734623",
  childWindow: true,
  returnUrl: "http://localhost/api/pgresponse/modal/redirect",
  retryCount: 3
}
```

Parameter	Description	Classification
merchantId	merchant id provided by BillDesk	mandatory
mandateTokenId	As received in <i>Create Mandate Token API</i> response	mandatory

authToken	As received in <i>Create Mandate Token API</i> response	mandatory
childWindow	true	mandatory
returnUrl	Merchant return url ((If this parameter is passed, the merchant will receive a redirection on this URL on payment completion, otherwise the merchant page will receive a function callback for childwindow))	optional
retryCount	count for retry (default count: 3)	optional

6.3. Modify/Delete Mandate Setup SDK

Prepare the flow config object

```
var modifyMandateFlowConfig = {
  merchantId: "BDMERCID",
  mandateTokenId: "TSFFDHWE",
  action: "modify",
  authToken: "OToken FDD5C104249A4CDAE734623",
  childWindow: true,
  returnUrl: "http://localhost/api/pgresponse/modal/redirect",
  retryCount: 3
}
```

Parameter	Description	Classification
merchantId	merchant id provided by BillDesk	mandatory
mandateTokenId	As received in <i>Update Mandate Token API</i> response	mandatory
authToken	As received in <i>Update Mandate Token API</i> response	mandatory
childWindow	true	mandatory
returnUrl	Merchant return url ((If this parameter is passed, the merchant will receive a redirection on this URL on payment completion, otherwise the merchant page will receive a function callback for childwindow))	optional
retryCount	count for retry (default count: 3)	optional

Prepare the responseHandler

```
var responseHandler = function (txn) {
  console.log("callback received status:: ",txn.status)
  console.log("callback received response:: ",txn.response)
}
```

Prepare the config object


```
var config = {
  responseHandler: responseHandler,
  merchantLogo: "data:image/png;base64:eqwewqesddhgjdxsc=="
  flowConfig: mandate_flow_config,
  flowType: "emandate"
}
```

Parameter	Description	Classification
responseHandler	JavaScript function to receive a transaction status object as callback to indicate completion of the transaction journey with required details	optional
merchantLogo	merchant's logo in base64 format	optional
flowConfig	Flow specific config	mandatory
flowType	payments/emandate/modify_mandate (default: payments)	optional

Finally pass the config object created to the below function to launch the SDK

```
window.loadBillDeskSdk(config)
```

After completing the transaction on SDK, the encrypted transaction response is provided to the merchant on their Return URL (ru).

Sample Encrypted Transaction Response:

```
{"terminal_state":"Y","orderid":"yukkq8hnxcrztea8op24bk","transaction_response":"eyJ4NXQjUz
l1Ni6l6mZlNllsbHUtQXB2Q1d0OXQwQ2VPdG5dkJ1VFk5aG1aSVZYWEJraFZBYkkiLCJraWQiOiJ0ZW
NoYnVsbHMlLCJhbGciOiJQUzI1NiJ9.ZXlKNE5YUWpVekkxTmlJNklqZHVUM1Z3TnpKdFVuUlVjMUIz
V2tzeFIVVmxSMU5HY1ZCQlVUZDBWRlJQWHpCeFUyeDVWRXhHVZVZVaUxDSnJhV1FpT2lKMFPXT
m9ZblZzYkhNaUxDSmxibU1pT2lKQk1USTRSME5OSWI3aVlXeG5Jam9pVWxOQkxVOUJSVkfOTWp
VMklUMC5hcGNyRfIZTFBreUt4YkxVWm1KR0pUcm10MTRGbWQ2MnNLLU5mYW04dXU5bEE2eU
w1QnBWcFNWRHV3Vmp1WHJwaUMySIBLQ2RNNTI2aGhCZ3lqTzQyVHJJd3BXcFpIZExpX0twODh
vdDhFcGtCVkZNVlRvcVA0RW5BaElFd0VUT0I2RnE5UlplRHRVheUlzODYxSlIXTGvqbzh3MTFGWWx
hZ2tJUGFLZlNLOG92QXlUUGNRTEtkVG1kOvAYNUZzRIF4QWJnS09pZDlMa1NWajBxVUhVRURsb0
5GSEYyVVdPNjVBdWZ0RXEYnWZEXQpvQUJHYW50UkhjMzIRdTI0Q1ZCekwwc0pXZ0M2OEdLWm
ZzZkVxaTc5a203Vm9kT0hqRFdmb1RfQkdJa2hTZ210SXlpUmhiZzAzNGgwcy1GRtBnb2UxTGZnLU
N4UW5BYXRqamFaUnZGeS1LNy01MUpYa0Z2UjJIRlV2cHNrcm1lV01vU1R0aVUya2pGclJpcllyZnJt
eGU4TFI3VETnZmVXRfNCX1gtdTjsa2g0UDNoMXFUY196TU9PTWRFbUNPNGNoNmCzS3k3VkJrc
GZvak5JSHZrcFhsenNmZkxpVlg3a0xqVzcRzhDcUlCT2lPODBYb0VHUUNfbno4bIBIQjM4STNQeXZ
WdjE1VDMtNHFiNF9nZ05FQ0JQTHlLcm9SWHU5QzRYaXZRBhdHd3liUkFVRUpYQ2ZiY0JxbUpOaX
pJRdY1b1JDZ1lCdDdGZVE3QlFaTVh4cUV4eDVQa0JVSmxPd2o3bS02Vm1aS25WcnV6cUxKZ3YtVVZ
```

5OFBIY01rZXpPRXVfQmJiUnZzZkRnS19TaTdSU2lBR0ZiWVFYSTBNMGs0TEdkbmVNWjlEVWlKTDAX
RXlpMmp0WUZZejZ6MC50Z3UxSEtCQjE2c2laVmc4LnRvRGVnUkl5aW1KUzJZYkxuVm4zclBtemNZ
SUNWNHRwUzRMZV9MQUHJVUZneHJtdU9JVE1EdVFWdVBMLWxyaktLT3RLb0VqZU41bEpqZGp2
T1VqdS1zVXJBMVpmanpjUF9tVHotYWRMb1c3YklMaThPS3BRLS1NdUZDcHBaRFBDbVdZemk0RH
c0YkdDenlhbzNKLtJCcFhRUzhfbXlrUHAxSlpZ2JSajJwYm5kd3g1cGdiRzg3SENjTXVBeXI1VktDMzVIZ
25DV2hBRzZkOFJOTHE5RDIgVnJJeTBvSlZveU5iaktLLXlKWmF2STNSRTIsTFNXMURCTnZhejBPd0xE
ak1GeG85YIVGMUZKc2FQR05zdWVxS2Y0OGh5dU9uREllb01RdWpmQjNKYIMteFIMTE43czJDQO
QxNIJKOEdYcVVIQ3FUM3NVVGFWa2ppV3p0NjcydzJZeHAYSkdQcmluckRreWNmZ0toUUFNcmdJU
WZqZWxReHdKdDU5YTY1dnBSc2pQOXpYOW9KYUhhMnBST3JHQ3BuWjNod1FDOXlKMmpKTEp5
WjBEU1pYUThwYkt4aDF2b3BLSks3amk1bGlwTFJ1ZTdwSExuWFdaNjJUV0k4MTdra1VyBThqUEVC
YjZ2cjJxUGRMMWVrTnVpUEpPSVhMRy1qZElyUFRfU1owb0FIN0VjbUY5NWlPb0tLZU0wTF9xeTFI
UERvX2Z6a1liYjRaLWVERm16Tjk4SlFnWkQwb0xpQmVCRS1lZXd3QUZrc3Zyald3a19FMmJRLVg0b
HMzNS1wOHM2bjc3NUtvYldmdmRJE52bVRvV20yZFRDMmVWLTJEZHItUnNZc3RWTXphdEgxb0
11ZVpZSkRaU1RSbzBsZHNmNENONEhqOURiVmtkSkJVNGxEUzA4NEx1SVNIYVNIUmJqOWpDRnV
GRWdaaWhlcOpUaWJTUhdhTmdUNUk0RUt0aUctWWQ0NkJGelg1M2I5c2hmMnlwLWZlM2RBU
HlXSkpuaWNSQnZxaWFuUEhzOFpVYlZnNEYwTktoTnRIWFNQdUR5OWIRUWJKT1NHT3pGaFJ3WX
pydVotbmg5cFkwWEw5aXJLWHD4anB0UEs0NEduSFUyV2J4bVhPWHRqb2tfQXo2XzN5S2w0NUtV
WS4xeGxjTVJyMjViRlIBSTRIMXpXTDZ3.bvO37G7KsjhCrB7geqW8dXgWXuhZRVxMfCsOKhFNI_eHp
ydxNnMEaAUNJ__-K-zk-jZTnfitCuol-PvkmaLCOycN_J1xWujvdsNT1gVedr53PS-
NUMLNaf2a8Wa9QiGWbIN8quxjXVIFkp_NiUcyepLdVi3lcsWOC1K1qjy2tiRgRN0YNDGW8KLr6dS97
7UErhbhJGNW0ggvE06PzTU_yYIGj7rxBx1wqkhLT53AooV05gv8Y5s8iV_tcr410w64o0fcrcuGU1R
mrufVFEnZNYZEAzCcxeKKvlg4xcZZDPrCJM_elzHBVSm_72U1ilBT-
3f8y7eRJD1V7aB_hqXrw","return_url":"https://www.demourl.com/sdk/uat/api/pgresponse/mo
dal/redirect"}

Decrypted Transaction response:

```
{
  "objectid": "transaction",
  "mercid": "BDMERCID",
  "transaction_date": "2020-11-20T13:55:59+05:30",
  "surcharge": "2.00",
  "payment_method_type": "netbanking",
  "amount": "300.00",
  "ru": "https://example.com/pgresponse/modal/redirect",
  "orderid": "yukkq8hnxcrztea8op24bk",
  "transaction_error_type": "success",
  "discount": "0.00",
  "transactionid": "U1230000041968",
  "txn_process_type": "nb",
  "bankid": "123",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2"
  },
  "itemcode": "DIRECT",
  "transaction_error_code": "TRS0000",
  "currency": "356",
}
```

```
"auth_status":"0300",
"transaction_error_desc":"Transaction Successful",
"charge_amount":"302.00"
}
```

Decrypted Transaction + Mandate Response

```
{
  "objectid":"transaction",
  "mercid":"BDMERCID",
  "transaction_date":"2020-12-09T20:23:41+05:30",
  "surcharge":"0.00",
  "payment_method_type":"card",
  "amount":"300.00",
  "mandate":{
    "mercid":"BDMERCID",
    "end_date":"2023-12-09",
    "payment_method_type":"card",
    "amount":"200.00",
    "subscription_desc":"Family",
    "createdon":"2020-12-09T20:23:41+05:30",
    "frequency":"adho",
    "customer_refid":"cust2092022465L5SB",
    "amount_type":"maximum",
    "mandateid":"MA1B7B00043026",
    "subscription_refid":"Sub209202246LP7QF",
    "currency":"356",
    "objectid":"mandate",
    "start_date":"2020-12-09",
    "status":"active"
  },
  "ru":"https://www.samplurl.com/sdk/uat/api/pgresponse",
  "orderid":"qq1leqh1cxnnpn1016x8",
  "transaction_error_type":"success",
  "discount":"0.00",
  "eci":"05",
  "bank_ref_no":"123456",
  "transactionid":"UPGM0000045450",
  "txn_process_type":"3ds",
  "bankid":"PGM",
  "additional_info":{
    "additional_info1":"Details1",
    "additional_info2":"Details2"
  },
  "itemcode":"DIRECT",
  "transaction_error_code":"TRS0000",
  "currency":"356",
  "auth_status":"0300",
  "transaction_error_desc":"Transaction Successful",
  "card":{
    "masked_value":"459150xxxxxx0055"
  },
  "charge_amount":"300.00"
}
```

Decrypted Only Mandate Setup Response (Active Status)

```
{
  "objectid": "mandate",
  "mercid": "DEMO",
  "end_date": "2021-05-24",
  "payment_method_type": "card",
  "amount": "1.00",
  "ru": "https://www.billdesk.com/sdk/uat/api/mandateresponse",
  "subscription_desc": "Term insurance",
  "verification_error_type": "success",
  "createdon": "2020-12-24T11:03:00+05:30",
  "frequency": "mnth",
  "customer_refid": "cust224053153XBO2L",
  "amount_type": "maximum",
  "additional_info": {},
  "mandateid": "MN00C603064226",
  "verification_error_code": "TRS0000",
  "subscription_refid": "Sub224053153QH785",
  "currency": "356",
  "card": {
    "paymentaccountid": "PA03E502694942",
    "masked_value": "xxxxxxxxxxxx0035"
  },
  "verification_error_desc": "Transaction Successful",
  "start_date": "2020-12-24",
  "status": "active",
  "customer": {}
}
```

Decrypted Only Mandate Setup Response (Deleted Status)

```
{
  "objectid": "mandate",
  "transactionid": "MT210930004056",
  "mandateid": "MA0FAF00067002",
  "mercid": "BDMERCID",
  "customer_refid": "cust930092049QPLFB",
  "subscription_refid": "Sub930092049MT8CD",
  "subscription_desc": "Term insurance",
  "start_date": "2021-10-01",
  "end_date": "2030-12-12",
  "amount": "200.00",
  "amount_type": "maximum",
  "currency": "356",
  "frequency": "mnth",
  "status": "deleted",
  "createdon": "2021-09-30T14:51:56+05:30",
  "deletedon": "2021-09-30T14:51:56+05:30",
  "payment_method_type": "card",
  "ru": "https://billdesk.techbulls.com/sdk/uat/api/pgresponse/modal/redirect",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2"
  }
}
```

```

},
"card":{
  "paymentaccountid":"PA0B8700048272",
  "masked_value":"xxxxxxxxxxxx0055",
  "holder_name":"test"
},
"customer":{

},
"verification_error_code":"TRS0000",
"verification_error_type":"success",
"verification_error_desc":"Transaction Successful",
"issuer_ref_no":"VcbjVMY500",
"recurrence_rule":"on",
"debit_day":"6"
}

```

7. INVOICE

7.1. The invoice object

Below are the details of the invoice object:

Attributes	Type	Description
objectid	String	String representing the object's type. Objects of the same type share the same value. This value will be fixed as invoice
invoice_id	String	Unique id assigned by BillDesk for each invoice
invoice_number	String	Unique number assigned by merchant for the invoice
invoice_display_number	String	Invoice number to be displayed to the customer
mercid	String	Unique mercid assigned by BillDesk to each merchant
status	String	Status of the invoice. Possible values are unpaid, rejected, stopped
debit_request_no	String	Sequence of the recurring charge. For e.g. Say a customer is on a monthly payment plan – the first recurring charge has the value as “1”, the second as “2” and so on. If the merchant does not process a debit say for the second month and then in the next (third) month when initiating a charge, this value should be set as “3” in the request.
customer_refid	String	Unique customer identifier as per merchant.
subscription_refid	String	Unique reference assigned by merchant for the subscription. Mandatory for recurring payment
subscriptionid	String	Unique subscriptionid created by BillDesk at the time of subscription generation
invoice_date	Date	Invoice generation Date
duedate	Date	Invoice due Date
debit_date	Date	Date on which customer will get the SI debit charge request
amount	String	Invoice amount
early_payment_duedate	String	Early payment date for invoice

early_payment_discount	String	Discount amount for early payment of invoice
early_payment_amount	String	Amount after adjusting any discount applicable.
late_payment_charges	String	Late payment charges if any
late_payment_amount	String	Payment amount after due date
net_amount	String	Current payable amount, adjusted for discount or late payment charges as applicable
currency	String	ISO currency code
verification_error_code	String	Represents the error code for a request with rejected status
verification_error_type	String	Represents the standard error category for a request with rejected status
verification_error_desc	String	Represents the description of the error code for a request with rejected status
mandateid	String	Unique mandateid generated by BillDesk for the associated mandate
description	String	Brief description for the invoice
createdon	timestamp	Invoice creation date and time format YYYY-MM-DDThh:mm:ssTZD
deletedon	timestamp	Invoice deletion date and time format YYYY-MM-DDThh:mm:ssTZD
rejectedon	timestamp	Invoice rejection date and time format YYYY-MM-DDThh:mm:ssTZD

7.2. Create Invoice

Request

Attributes	Required/ Optional/Conditional	Description
invoice_number	required	Unique number assigned by merchant for the invoice
invoice_display_number	required	Invoice number to be displayed to the customer
mercid	required	Unique mercid assigned by BillDesk to each merchant
customer_refid	required	Unique customer identifier as per merchant.
additional_info	optional	Array of 10 additional_info values that can be attached to the transaction
subscription_refid	required	Unique reference assigned by merchant for the subscription. Mandatory for recurring payment
subscriptionid	optional	Unique subscriptionid created by BillDesk at the time of subscription generation
debit_request_no	conditional	Sequence of the recurring charge. For e.g. Say a customer is on a monthly payment plan – the first recurring charge has the value as “1”, the second as “2” and so on. If the merchant does not process a debit say for the second month and then in the next (third) month when initiating a charge, this value should be set as “3” in the request.

invoice_date	required	Invoice generation date
duedate	required	Invoice due date
debit_date	required	Date on which customer will get the SI debit charge request
amount	required	Invoice amount
early_payment_duedate	optional	Early payment date for invoice
early_payment_discount	optional	Discount amount for early payment of invoice
early_payment_amount	optional	Amount after adjusting any discount applicable.
late_payment_charges	optional	Late payment charges if any
late_payment_amount	optional	Payment amount after due date
net_amount	required	Current payable amount, adjusted for discount or late payment charges as applicable
currency	required	ISO currency code
mandateid	required	Unique mandateid generated by BillDesk for the associated mandate
description	required	Brief description for the invoice

Response

Returns the invoice object created.

(a) For card

Sample Request

```
curl https://pguat.billdesk.io/pgsi/v1_2/invoices/create \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
-H 'Accept: application/jose' \
-X POST \
-d '{
  "mercid": "BDMERID",
  "subscription_refid": "SUBS0920202006",
  "customer_refid": "CUREF000004",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2",
    "additional_info3": "Details3",
    "additional_info4": "Details4",
    "additional_info5": "Details5",
    "additional_info6": "Details6",
    "additional_info7": "Details7",
    "additional_info8": "Details8",
    "additional_info9": "Details9",
    "additional_info10": "Details10"
  },
  "invoice_number": "INVC00002",
  "invoice_display_number": "PLN000001",
  "invoice_date": "2020-10-03",
  "duedate": "2020-11-02",
}
```

```
"debit_date":"2020-11-02",
"amount":"500.00",
"early_payment_duedate":"2019-10-25",
"early_payment_discount":"15.00",
"early_payment_amount":"485.00",
"late_payment_charges":"50.00",
"late_payment_amount":"550.00",
"net_amount":"500.00",
"currency":"356",
"mandateid":"MA090F00042241",
"description":"Invoice for Oct 2020"
}'
```

Sample Response

```
{
  "invoice_number":"INVC00002",
  "objectid":"invoice",
  "invoice_display_number":"PLN000001",
  "mercid":"BDMERCID",
  "subscription_refid":"SUBS0920202006",
  "customer_refid":"CUREF000004",
  "invoice_date":"2020-10-03",
  "duedate":"2020-11-02",
  "debit_date":"2020-11-02",
  "amount":"500.00",
  "net_amount":"500.00",
  "early_payment_due_date":"2019-10-25",
  "early_payment_discount":"15.00",
  "late_payment_charges":"50.00",
  "late_payment_amount":"550.00",
  "currency":"356",
  "mandateid":"MA090F00042241",
  "description":"Invoice for Oct 2020",
  "status":"unpaid",
  "invoice_id":"IA24A200000942",
  "verification_error_code":"INP0000",
  "verification_error_type":"Invoice Successful",
  "verification_error_desc":"success",
  "additional_info":{
    "additional_info1":"Details1",
    "additional_info2":"Details2",
    "additional_info3":"Details3",
    "additional_info4":"Details4",
    "additional_info5":"Details5",
    "additional_info6":"Details6",
    "additional_info7":"Details7",
    "additional_info8":"Details8",
    "additional_info9":"Details9",
    "additional_info10":"Details10"
  },
  "createdon":"2020-11-02T21:51:19+05:30"
}
```


In case if any customer has opted for not allowing the invoice to get processed successfully from issuer; BillDesk will receive a callback from the issuer, and in turn BillDesk will notify the merchant via webhook notification with an invoice status as stopped.

(b) For *upi*

Sample Request

```
{
  "subscription_refid": "billhosted92004",
  "mercid": "BDMERCID",
  "customer_refid": "CUST143468004",
  "invoice_number": "MEINVOU202009240200",
  "invoice_display_number": "202009240200",
  "invoice_date": "2020-10-11",
  "duedate": "2020-10-15",
  "debit_date": "2020-10-13",
  "amount": "500.00",
  "debit_request_no": "1",
  "early_payment_duedate": "2019-10-04",
  "early_payment_discount": "15.00",
  "early_payment_amount": "485.00",
  "late_payment_charges": "50.00",
  "late_payment_amount": "550.00",
  "net_amount": "500.00",
  "additional_info": {
    "additional_info1": "Details1",
    "additional_info2": "Details2",
    "additional_info3": "Details3",
    "additional_info4": "Details4",
    "additional_info5": "Details5",
    "additional_info6": "Details6",
    "additional_info7": "Details7",
    "additional_info8": "Details8",
    "additional_info9": "Details9",
    "additional_info10": "Details10"
  },
  "currency": "356",
  "mandateid": "MA051200040823",
  "description": "Invoice"
}
```

Sample Response

```
{
  "invoice_id": "IA062C00000721",
  "invoice_number": "MEINVOU202009240200",
  "invoice_display_number": "202009240200",
  "mercid": "BDMERCID",
  "subscription_refid": "billhosted92004",
  "customer_refid": "CUST143468004",
  "invoice_date": "2020-10-11",
  "duedate": "2020-10-15",
  "debit_date": "2020-10-13",

```

```

"amount": "500.00",
"debit_request_no": "1",
"early_payment_duedate": "2019-10-04",
"early_payment_discount": "15.00",
"early_payment_amount": "485.00",
"late_payment_charges": "50.00",
"late_payment_amount": "550.00",
"net_amount": "550.00",
"currency": "356",
"mandateid": "MA051200040823",
"description": "Invoice",
"status": "unpaid",
"verification_error_code": "INP0000",
"verification_error_type": "Invoice Successful",
"verification_error_desc": "success",
"additional_info": {
  "additional_info1": "Details1",
  "additional_info2": "Details2",
  "additional_info3": "Details3",
  "additional_info4": "Details4",
  "additional_info5": "Details5",
  "additional_info6": "Details6",
  "additional_info7": "Details7",
  "additional_info8": "Details8",
  "additional_info9": "Details9",
  "additional_info10": "Details10"
},
}

```

In case if any customer has opted for not allowing the invoice to get processed successfully from UPI PSP App; BillDesk will receive a callback from the acquirer, and in turn BillDesk will notify the merchant via webhook notification with an invoice status as stopped.

7.3. Retrieve Invoice

Merchants can retrieve invoice details by using **mercid** and **invoice_number**

Request

Attributes	Required/Optional	Description
mercid	Required	Unique identifier as defined by BillDesk for each merchant
invoice_number	Required	Unique number assigned by merchant for the invoice

Response

Returns the invoice object

Sample Request

```
curl https://pguat.billdesk.io/pgsi/vel_2/invoices/get \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
-H 'Accept: application/jose' \
-X POST \
-d '{
  "mercid":"BDMERCID",
  "invoice_number":"INVCABCUP10002"
}'
```

Sample Response

```
{
  "invoice_number":"INVCABCUP10002",
  "objectid":"invoice",
  "invoice_display_number":"DISINVCABCUP10002",
  "mercid":"BDMERCID",
  "subscription_refid":"SCREFUP00001",
  "customer_refid":"CSREFUP00001",
  "invoice_date":"2021-05-10",
  "duedate":"2021-05-18",
  "debit_date":"2021-05-16",
  "amount":"500.00",
  "net_amount":"500.00",
  "early_payment_due_date":"2021-05-12",
  "early_payment_discount":"10.00",
  "early_payment_amount":"490.00",
  "late_payment_charges":"20.00",
  "late_payment_amount":"520.00",
  "currency":"356",
  "mandateid":"MA1F8A00050242",
  "description":"Invoice for May 2021",
  "status":"unpaid",
  "invoice_id":"IA047F00003663",
  "verification_error_code":"INS0000",
  "verification_error_type":"Invoice Successful",
  "verification_error_desc":"success",
  "additional_info":{
    "additional_info1":"Details1",
    "additional_info2":"Details2",
    "additional_info3":"Details3",
    "additional_info4":"Details4",
    "additional_info5":"Details5",
    "additional_info6":"Details6",
    "additional_info7":"Details7",
    "additional_info8":"Details8",
    "additional_info9":"Details9",
    "additional_info10":"Details10"
  },
  "createdon":"2021-05-16T19:20:08+05:30",
  "debit_request_no":"1"
}
```

8. CREATE TRANSACTION (SI) API

Merchant can use this API to initiate a recurring charge.

Request

Attributes	Required/ Optional	Description
orderid	Required	Unique orderid generated by merchant to identify the transaction
mercid	Required	Unique identifier as defined by BillDesk for each merchant
amount	Required	Transaction amount in two decimals
currency	Required	ISO currency of the transaction amount
txn_process_type	Required	Fixed value si
mandateid	Required	Mandate id created by BillDesk at the time of recurring setup. Mandatory for recurring payment
subscription_refid	Required	Unique reference assigned by merchant for the subscription. Mandatory for recurring payment
invoice_id	Required	Unique invoice id generated by BillDesk
debit_request_no	Conditional	Sequence of the recurring charge. For e.g. Say a customer is on a monthly payment plan – the first recurring charge has the value as “1”, the second as “2” and so on. If the merchant does not process a debit say for the second month and then in the next (third) month when initiating a charge, this value should be set as “3” in the request.
additional_info	Optional	Array of 10 additional_info values that can be attached to the transaction
itemcode	Required	Itemcode value as provided by BillDesk, with a default value DIRECT
settlement_lob	Optional	Settlement line of business preconfigured by BillDesk for funds settlement to merchant account
customer	Optional	Customer object

Response

Returns the transaction object.

Sample Request

```
curl https://pguat.billdesk.io/payments/v1_2/transactions/create\
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
```

```
-H 'Accept: application/jose' \  
-X POST \  
-d '{  
  "mercid":"BDMERCID",  
  "orderid":"RECCH000011",  
  "amount":"200.00",  
  "currency":"356",  
  "bankid":"456",  
  "itemcode":"DIRECT",  
  "additional_info":{  
    "additional_info1":"Details1",  
    "additional_info2":"Details2",  
    "additional_info3":"Details3",  
    "additional_info4":"Details4",  
    "additional_info5":"Details5",  
    "additional_info6":"Details6",  
    "additional_info7":"Details7",  
    "additional_info8":"Details8",  
    "additional_info9":"Details9",  
    "additional_info10":"Details10"  
  },  
  "txn_process_type":"si",  
  "mandateid":"MA090F00042241",  
  "invoice_id":"IA24A200000942",  
  "subscription_refid":"SUBS0920202006"  
}'
```

Sample Response

```
{  
  "objectid":"transaction",  
  "transactionid":"U4560000039473",  
  "orderid":"RECCH000011",  
  "mercid":"BDMERCID",  
  "transaction_date":"2020-11-04T01:23:04+05:30",  
  "amount":"200.00",  
  "surcharge":"0.00",  
  "discount":"0.00",  
  "charge_amount":"200.00",  
  "currency":"356",  
  "additional_info":{  
    "additional_info1":"Details1",  
    "additional_info2":"Details2",  
    "additional_info3":"Details3",  
    "additional_info4":"Details4",  
    "additional_info5":"Details5",  
    "additional_info6":"Details6",  
    "additional_info7":"Details7",  
    "additional_info8":"Details8",  
    "additional_info9":"Details9",  
    "additional_info10":"Details10"  
  },  
  "txn_process_type":"si",  
  "bankid":"456",  
}
```

```
"itemcode":"DIRECT",
"bank_ref_no":"123456",
"auth_status":"0300",
"transaction_error_code":"TRS0000",
"transaction_error_desc":"Transaction Successful",
"transaction_error_type":"success",
"authcode":"123456",
"eci":"02",
"payment_method_type":"card",
"card":{
  "masked_value":"459150xxxxxx0055"
},
"mandateid":"MA090F00042241"
}
```

*An additional attribute *'debit_request_no'* needs to be sent in case of a UPI recurring change. Hence, refer the sample for UPI recurring charge below:

Sample Request

```
{
  "mercid":"BDMERCID",
  "orderid":"order1017",
  "amount":"599.00",
  "currency":"356",
  "itemcode":"DIRECT",
  "device":{
    "init_channel":"internet",
    "ip":"124.124.1.1",
    "mac":"11-AC-58-21-1B-AA",
    "imei":"990000112233445",
    "user_agent":"Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0",
    "accept_header":"text/html"
  },
  "customer":{
    "first_name":"John",
    "last_name":"Doe",
    "mobile":"9800000000",
    "mobile_alt":"9800000000",
    "email":"john.doe@somedomain.com",
    "email_alt":"john.doe@somedomain.com"
  },
  "mandateid":"MA221100044350",
  "debit_request_no":"1",
  "subscription_refid":"Sub229141820ML0FM",
  "txn_process_type":"si"
}
```

Sample Response

```
{
  "objectid":"transaction",

```

```

"transactionid":"T1010008590164",
"orderid":"order1017",
"mercid":"BDMERCID",
"transaction_date":"2020-12-30T11:57:40+05:30",
"amount":"599.00",
"surcharge":"0.00",
"discount":"0.00",
"charge_amount":"599.00",
"currency":"356",
"additional_info":{
  "additional_info1":"Details1",
  "additional_info2":"Details2",
  "additional_info3":"Details3",
  "additional_info4":"Details4",
  "additional_info5":"Details5",
  "additional_info6":"Details6",
  "additional_info7":"Details7",
  "additional_info8":"Details8",
  "additional_info9":"Details9",
  "additional_info10":"Details10"
},
"customer":{
  "first_name":"John",
  "last_name":"Doe",
  "mobile":"9800000000",
  "mobile_alt":"9800000000",
  "email":"john.doe@somedomain.com",
  "email_alt":"john.doe@somedomain.com"
},
"txn_process_type":"si",
"itemcode":"DIRECT",
"auth_status":"0300",
"bank_ref_no":"123123",
"transaction_error_code":"TRS0000",
"transaction_error_desc":"Transaction Successful",
"transaction_error_type":"success",
"payment_method_type":"upi",
"mandateid":"MA221100044350",
"debit_request_no":"1"
}

```

9. RETRIEVE TRANSACTION API

Merchant can retrieve the transaction either by using **mercid** and **orderid** or **transactionid**

Attributes	Mandatory/ Optional/ Conditional	Description
orderid	Conditional	Unique orderid generated by Merchant to identify the transaction

transactionid	Conditional	Unique transactionid generation by BillDesk for the transaction
mercid	Mandatory	Unique identifier as defined by BillDesk for each merchant

Response

Returns the transaction object.

Sample Request

```
curl https://pguat.billdesk.io/payments/vel_2/transactions/get \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
-H 'Accept: application/jose' \
-X POST \
-d '{
  "mercid":"BDMERCID",
  "orderid":" ORDER28102020005"
}
```

Sample Response

```
{
  "objectid":"transaction",
  "transactionid":"U1230000038061",
  "orderid":"ORDER28102020005",
  "mercid":"BDMERCID",
  "transaction_date":"2020-10-28T01:03:56+05:30",
  "amount":"500.00",
  "surcharge":"0.00",
  "discount":"0.00",
  "charge_amount":"500.00",
  "currency":"356",
  "additional_info":{
    "additional_info1":"Details1",
    "additional_info2":"Details2",
    "additional_info3":"Details3",
    "additional_info4":"Details4",
    "additional_info5":"Details5",
    "additional_info6":"Details6",
    "additional_info7":"Details7",
    "additional_info8":"Details8",
    "additional_info9":"Details9",
    "additional_info10":"Details10"
  },
  "ru":"https://www.testmerchant.com/abcpage",
  "customer":{
    "first_name":"John",
    "last_name":"Doe",
    "mobile":"9800000000",
    "mobile_alt":"999955555",
    "email":"john.doe@somedomain.com",
  }
}
```



```

    "email_alt": "john.alt@somedomain.com"
  },
  "txn_process_type": "nb",
  "bankid": "123",
  "itemcode": "DIRECT",
  "auth_status": "0300",
  "transaction_error_code": "TRS0000",
  "transaction_error_desc": "Transaction Successful",
  "transaction_error_type": "success",
  "payment_method_type": "netbanking"
}

```

10. CREATE REFUND

Merchant can initiate a refund against a transaction.

Attributes	Mandatory/ Optional/ Conditional	Description
transactionid	Mandatory	Original transactionid generated by BillDesk for the transaction
orderid	Mandatory	Original transaction orderid generated by merchant to identify the transaction
mercid	Mandatory	Unique identifier as defined by BillDesk for each merchant
transaction_date	Mandatory	Original BillDesk transaction date and time in YYYY-MM-DDThh:mm:ssTZD format
txn_amount	Mandatory	Original transaction amount in two decimals
refund_amount	Mandatory	Refund amount in two decimals
currency	Mandatory	ISO currency of the transaction amount
device	Optional	Device object
merc_refund_ref_no	Mandatory	Merchant generated unique reference number for refund transaction

Returns the refund object.

Sample request

```

curl https://pguat.billdesk.io/payments/v1_2/refunds/create \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
-H 'Accept: application/jose' \
-X POST \
-d '{
  "transactionid": "U1230000038061",
  "orderid": "ORDER28102020005",
  "mercid": "BDMERCID",
  "transaction_date": "2020-10-28T01:03:56+05:30",
  "txn_amount": "500.00",

```

```

"refund_amount":"250.00",
"currency":"356",
"merc_refund_ref_no":"NBREF0001",
"device":{
  "deviceid":"eyJhbGciOiJSUzUxPj93pedfaefdawefWEFwefWAF",
  "app_id":"SAMPLEAPP001",
  "init_channel":"internet",
  "ip":"124.124.1.1",
  "mac":"11-AC-58-21-1B-AA",
  "imei":"990000112233445",
  "user_agent":"Mozilla/5.0 (Windows NT 10)",
  "accept_header":"text/html",
  "fingerprintid":"61b12c18b5d0cf901be34a23ca64bb19"
}
}

```

Sample response

```

{
  "objectid":"refund",
  "refundid":"U12300000380611",
  "transactionid":"U1230000038061",
  "orderid":"ORDER28102020005",
  "mercid":"BDMERCID",
  "transaction_date":"2020-10-28T01:03:56+05:30",
  "txn_amount":"500.00",
  "refund_amount":"250.00",
  "currency":"356",
  "refund_date":"2020-11-03T18:11:45+05:30",
  "merc_refund_ref_no":"NBREF0001",
  "refund_status":"0799"
}

```

11. RETRIEVE REFUND

Merchant can retrieve refund details either by using **mercid** and **merc_refund_ref_no** or **refundid**

Attributes	Mandatory/ Optional/ Conditional	Description
mercid	Mandatory	Unique identifier as defined by BillDesk for each merchant
refundid	Conditional	Unique refundid generated by BillDesk
merc_refund_ref_no	Conditional	Merchant generated unique reference number for this refund transaction

Response

Returns the refund object.

Sample Request

```
curl https://pguat.billdesk.io/payments/vel_2/refunds/get \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
-H 'Accept: application/jose' \
-X POST \
-d '{
  "mercid":"BDMERCID",
  "merc_refund_ref_no":"REFTST00001"
}'
```

Sample Response

```
{
  "objectid":"refund",
  "refundid":"U45600000256681",
  "transactionid":"U4560000025668",
  "orderid":"ORDERCR00001230002",
  "mercid":"BDMERCID",
  "transaction_date":"2020-09-18T01:08:41+05:30",
  "txn_amount":"500.00",
  "refund_amount":"134.00",
  "currency":"356",
  "refund_date":"2020-09-18T01:38:29+05:30",
  "merc_refund_ref_no":"REFTST00001",
  "refund_status":"0799",
  "refund_arn":"5678",
  "bank_refund_date":"2020-09-25T01:12:29+05:30"
}
```

12. RETRIEVE MANDATE

Retrieve details of a single mandate as per the *mandateid*.

Request

Attributes	Required/Optional	Description
mandateid	Conditional	Unique mandateid created by BillDesk

Response

Returns the mandate object.

Sample Request

```
curl https://pguat.billdesk.io/pgsi/vel_2/mandates/get \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
```

```
-H 'Content-Type: application/jose' \  
-H 'Accept: application/jose' \  
-X POST \  
-d '{  
    "mandateid": "MA205800042329"  
}'
```

Sample Response

```
{  
  "objectid": "mandate",  
  "mandateid": "MA205800042329",  
  "mercid": "BDMERCID",  
  "customer_refid": "CUREFUPI01112020004",  
  "subscription_refid": "SUBSUPI01112020004",  
  "subscription_desc": "Unlimited Plan",  
  "start_date": "2020-11-03",  
  "end_date": "2021-10-10",  
  "amount": "500.00",  
  "amount_type": "maximum",  
  "currency": "356",  
  "frequency": "adho",  
  "status": "active",  
  "createdon": "2020-11-03T21:59:28+05:30",  
  "payment_method_type": "upi",  
  "additional_info": {  
    "additional_info1": "Details1",  
    "additional_info2": "Details2",  
    "additional_info3": "Details3",  
    "additional_info4": "Details4",  
    "additional_info5": "Details5",  
    "additional_info6": "Details6",  
    "additional_info7": "Details7",  
    "additional_info8": "Details8",  
    "additional_info9": "Details9",  
    "additional_info10": "Details10"  
  },  
  "upi": {  
    "paymentaccountid": "PA0EAD00025941",  
    "vpa": "testvpa@abc",  
    "psp": "abc"  
  },  
  "customer": {  
    "first_name": "John",  
    "last_name": "Doe",  
    "email": "john.doe@somedomain.com",  
    "email_alt": "john.doe@somedomain.com",  
    "mobile": "9800000000",  
    "mobile_alt": "9800000000"  
  }  
}
```

13. LIST ALL MANDATES

Retrieve all mandates created for a customer or based on created date range [not exceeding 7 days].

Request

Attributes	Required/Optional	Description
Mercid	Required	Unique identifier as defined by BillDesk for each merchant
customer_refid	Conditional	Unique reference of the customer in the merchant system, and can be linked to different subscriptions of the same customer.
subscription_refid	Conditional	Unique reference for the subscription in the merchant system
from_date	Optional	Start mandate creation date for the date range.
to_date	Optional	End mandate creation date for the date range.

Response

Returns list of all mandates associated with the customer or for the given date range.

Sample Request

```
curl https://pguat.billdesk.io/pgsi/vel 2/mandates/list \
-H 'BD-Traceid: AT0001' \
-H 'BD-Timestamp: 1544508848' \
-H 'Content-Type: application/jose' \
-H 'Accept: application/jose' \
-X POST \
-d '{
  "mercid":"BDMERCID",
  "customer_refid":"CUST123456"
}'
```

Sample Response

```
{
  "objectid":"list",
  "mandates":[
    {
      "objectid":"mandate",
      "mandateid":"BDMD000001",
      "mercid":"BDMERCID",
      "customer_refid":"CUST123456",
      "subscription_refid":"abcd12344567",
      "subscription_desc":"Unlimited Plan",
    }
  ]
}
```

```

    "start_date": "2019-12-22",
    "end_date": "2022-12-22",
    "amount": "500.00",
    "amount_type": "max",
    "currency": "356",
    "frequency": "adho",
    "status": "active",
    "createdon": "2019-12-22T11:26:14+05:30",
    "payment_method_type": "card",
    "card": {
      "paymentaccountid": "pay323232",
      "masked_value": "xxxxxxxxxxxx2222",
    },
    "customer": {
      "first_name": "John",
      "last_name": "Doe",
      "mobile": "9800000000",
      "mobile_alt": "9800000000",
      "email": "john.doe@somedomain.com",
      "email_alt": "john.doe@somedomain.com"
    }
  },
  {
  }
]
}

```

14. OBJECTS

14.1. transaction

Attributes	Type	Description
objectid	String	String representing the object's type. This value will be fixed as - transaction
transactionid	String	Unique transactionid generation by BillDesk for the transaction
orderid	String	Unique orderid generated by merchant to identify the transaction
mercid	String	Unique identifier as defined by BillDesk for each merchant
transaction_date	timestamp	BillDesk transaction date and time in YYYY-MM-DDThh:mm:ssTZD format
amount	String	Transaction amount in two decimals
surcharge	String	Customer surcharge in two decimals applied to the transaction amount, if any
discount	String	Customer discount in two decimals applied to the transaction amount, if any
charge_amount	String	Total charge to the customer

currency	number	ISO currency of the transaction amount
txn_process_type	String	Applicable where the payment_method_type = card , and will take 3ds as the default
bankid	String	BillDesk defined unique identifier for bank or acquirer
ru	String	Merchant return url
additional_info	Object	Array of 10 additional_info values that can be attached to the transaction
itemcode	String	Itemcode value as provided by BillDesk, with a default value DIRECT
bank_ref_no	String	Transaction reference number generated by bank or acquirer
auth_status	String	Represents the authorization status of the transaction with the following possible values: • 0300 - transaction is successful • 0002 - transaction is pending for authorization • 0399 - transaction failed
settlement_lob	String	Settlement line of business preconfigured by BillDesk for funds settlement to merchant account
customer	Object	Customer object
device	Object	Device object
transaction_error_code	String	Represents the error code for a transaction with 0399 status
transaction_error_type	String	Represents the standard error category for a transaction with 0399 status
transaction_error_desc	String	Represents the description of the error code for a transaction with 0399 status
authcode	String	Authorization code received from the acquirer for a successfully authorized card transaction
eci	String	eci value for the authentication taken for the card transaction
payment_method_type	String	Represents the method of payment e.g. card , upi , netbanking , wallet . The associated object will also be attached to the transaction
card	Object	Payment method object (applicable when payment method is card)
customer_refid	String	Unique customer identifier as per merchant
links	Object	Associated links with the object

14.2. customer

Attributes	Type	Description
first_name	String	Customer's First Name
last_name	String	Customer's Last Name
email	String	Customer's Email address
email_alt	String	Customer's alternate Email Address
mobile	String	Customer's Mobile number
mobile_alt	String	Customer's Alternate Mobile number

14.3. device

Attributes	Type	Mandatory/ Optional/ Conditional	Description
deviceid	String	Optional	Unique deviceid .
appid	String	Optional	Merchant appid.
accept_header	String	Optional	Customer browser accept header values
init_channel	String	Mandatory	Initiating channel, accepted values are internet , app or kiosk
ip	String	Mandatory	Customer's IP address
mac	String	Optional	Customer's mac address when the initiating channel is internet
imei	String	Optional	Customer's mobile IMEI when initiating channel is app
user_agent	String	Mandatory	Customer's browser user agent
accept_header	String	Optional	Customer's browser accept header values
fingerprintid	String	Optional	Unique device identifier

14.4. refund

Attributes	Type	Description
objectid	String	String representing the object's type. This value will be fixed as – refund
refundid	String	Unique refundid generation by BillDesk for the transaction
transactionid	String	Original transactionid generation by BillDesk for the transaction
orderid	String	Original transaction orderid generated by Merchant to identify the transaction
mercid	String	Unique identifier as defined by BillDesk for each merchant
transaction_date	timestamp	Original transaction date and time in YYYY-MM-DDThh:mm:ssTZD format
txn_amount	String	Original transaction amount in two decimals
refund_amount	String	Refund amount in two decimals
refund_status	String	Represents the refund status of the transaction with the following possible values: • 0799 - transaction is refunded • 0699 - transaction is cancelled
currency	String	ISO currency of the transaction amount
device	Object	Device object
refund_date	Timestamp	Refund date and time in YYYY-MM-DDThh:mm:ssTZD format
refund_arn	String	Refund arn number as received by bank after refund confirmation
merc_refund_ref_no	String	Merchant generated unique reference number for refund transaction
bank_refund_date	String	Refund processing date

14.5. mandate

Attributes	Type	Description
objectid	String	String representing the object's type. Objects of the same type share the same value. This value will be fixed as - mandate
mandateid	String	Unique id assigned to each successful mandate request by BillDesk
Mercid	String	Unique identifier as defined by BillDesk for each merchant
verification_type conditional	String	Indicates type of verification, accepted values are enach_dc, enach_nb, nb, upi or 3ds
customer_refid	String	Unique reference of the customer in the merchant system, and can be linked to different subscriptions of the same customer
subscription_refid	String	Unique reference assigned by merchant for the subscription
subscription_desc	String	Customer friendly description for the subscription
start_date	String	Subscription start date in yyyy-mm-dd format
end_date	String	Subscription end date in yyyy-mm-dd format
amount_type	String	Indicates whether subscription amount is fixed or has a maximum limit. The following values are possible. maximum - indicates the maximum amount that can be charged for the subscription, fixed – indicates the fixed amount to be charged for the subscription
amount	String	Subscription amount
currency	String	ISO currency code for the amount
frequency	String	Frequency at which the subscription debits will be processed. Possible values are * week - weekly * mnth - monthly * bimn – bi-monthly * qurt - quarterly * bian – bi-annually * year - yearly * adho – adhoc
additional_info	Array	Array of 10 additional_infos [additional_info1-additional_info10]. There are custom data fields for merchant
payment_method_type	String	Payment method type - allowed values are card , upi or bankaccount
card	Object	Represents the method of payment linked with the subscription

status	String	Represents the status of the mandate and depends upon the linked payment method. The following values are possible: card • initiated - mandate request has been initiated. This state only exists for billdesk hosted workflow • pending - mandate request has been initiated and a confirmation is awaited for the validation of the payment method • active - mandate is active and recurring charges can be processed • rejected - mandate has been rejected by the payment provider or issuer • deleted - mandate has been deleted by the customer or merchant upi • Initiated - mandate request has been initiated. This state only exists for billdesk hosted workflow • pending -mandate request has been initiated and a confirmation is awaited for the validation of the payment method • active - mandate is active and recurring charges can be processed • rejected - mandate has been rejected by the payment provider or issuer • deleted - mandate has been deleted by the customer or merchant bankaccount • initiated - mandate request has been initiated. This state only exists for billdesk hosted workflow • pending - mandate request has been sent for account verification • received - mandate request initial response has been received and a confirmation is awaited for the validation of the payment method • active – mandate is active and recurring charges can be processed • rejected – mandate has been rejected by the payment provider or issuer • deleted – mandate has been deleted by the customer or merchant
--------	--------	---

bank_umrn	String	Represents a unique mandate reference number, provided for 'upi' payment_method_types
ru	String	Merchant return url where the authentication led redirection response will be received
verification_error_code	String	Represents the error code for a mandate with rejected status
verification_error_type	String	Represents the standard error category for a mandate with rejected status
verification_error_desc	String	Represents the description of the error code for a mandate with rejected status
recurrence_rule	String	Applicable for UPI. Values can be on/ before/ after
debit_day	String	Value that is specified depending upon the frequency. The values are as follows: - For a Weekly frequency the value for this field must be between 1 to 7 where 1 means Monday, 2 is Tuesday etc. - Fortnightly (starting from 1 to 15/16 [depending on the month]) - Monthly/ Bi monthly/ Quarterly/ Half-yearly/ Yearly [1 – 30/31 (if 30/31 is not available, the last day of the month will be considered for the debit)].
createdon	Timestamp	Mandate creation date and time format YYYY-MM-DDThh:mm:ssTZD
deletedon	Timestamp	Mandate deletion date and time format YYYY-MM-DDThh:mm:ssTZD
rejectedon	Timestamp	Mandate rejection date and time format YYYY-MM-DDThh:mm:ssTZD
links	Object	Associated links with the object
customer	Object	Customer object association with the mandate
device	Object	Customer device details

14.5.1. Card

Attributes	Type	Description
paymentaccountid	String	Unique accountid created by BillDesk
masked_value	String	Masked card number, including the first six and the last four digits

14.5.2. UPI

Attributes	Type	Description
paymentaccountid	String	Unique accountid created by BillDesk
vpa	String	Virtual Payment Address of user
psp	String	Payment system player of the user

14.5.3. Bank Account

Attributes	Type	Description
paymentaccountid	String	Unique accountid created by BillDesk
type	String	Bank account type (savings / current)
masked_value	String	Masked Bank account number
holder_name	String	Bank account holder name.
bank_id	String	Customer's issuing bank

15. ERROR

BillDesk uses conventional HTTP response codes to indicate the success or failure of an API request. In general, codes in the 2xx range indicate success, codes in the 4xx range indicate an error that failed given the information provided (e.g., a required parameter was omitted, etc.), and codes in the 5xx range indicate an error with BillDesk servers (these are rare).

Not all errors map cleanly onto HTTP response codes, however. When a request is valid but does not complete successfully (e.g., a recharge validation with the operator has failed), we return a 422/200 error code.

15.1. The error Object

Below are the details of the error object

Attributes	Type	Description
Status	String	This will be same as HTTP response code
error_type	String	This indicates of the category of errors that could be encountered, for e.g. validation_error, api_connection_error, etc.
error_code	String	Under the error_type, there could be different error_codes that could be returned depending upon the API and the underlying interaction
message	String	More details as a description for the error_code

Sample Response

```
{  
  "status":422,  
  "error_type":"invalid_data_error",  
  "error_code":"TRIDE0011",  
}
```

```
"message": "Invalid orderid"
}
```

15.2. Standard HTTP response codes

Response code	Category	Description
200	OK	Request OK
400	Bad request	The request was invalid or cannot be otherwise served. An accompanying error message will explain further.
401	Unauthorized	Authentication credentials were missing or incorrect.
403	Forbidden	The request is understood, but it has been refused or access is not allowed. An accompanying error message will explain why.
404	Not Found	The URL requested is invalid or the resource requested, such as a user, does not exist. Also returned when the requested format is not supported by the requested method.
405	Method not allowed	The resource doesn't support the specified HTTP verb.
406	Not Acceptable	Returned by the Search API when an invalid format is specified in the request.
409	Conflict	Duplicate request. Traceid is duplicate or biller account already exists or there is a value present with the same key.
415	Unsupported Media Type	The server is refusing to service the request because the entity of the request is in a format not supported by the requested resource for the requested method
422	Unprocessable Entity	Validation Error
500	Internal server error	Internal processing error
502	Bad Gateway	Application down
504	Gateway timeout	Timeout in the application processing

15.3. Standard error types

Error type	Details
duplicate_request_error	Duplicate traceid received in the request
invalid_request_error	The request format is not correct
authentication_error	The authentication signature received in the request could not be matched
inaccessible_resource_error	The resource is forbidden and can't be accessed
not_found_error	The resource could not be found
invalid_method_error	Method in the request is not supported
invalid_media_type_error	Unspported media type in the request
invalid_data_error	The request could not be processed due to incorrect/inconsistent parameter values
api_validation_error	The request could not be processed due to business logic violation
api_processing_error	Some internal error occurred while processing the request at BillDesk

api_connection_error	There is a problem in connecting to internal applications
----------------------	---

15.4. Verification error types

Error type	Details
authentication_error	Payment provider could not validate the payment method
customer_cancel_error	Transaction verification cancelled by customer
api_validation_error	The request could not be processed due to business logic violation
api_processing_error	Some internal error occurred while processing the request at BillDesk
api_connection_error	There is a problem in connecting to internal applications

15.5. HATEOAS

Hypermedia As The Engine Of Application State (HATEOAS) is a constraint of the REST application architecture that distinguishes it from other network application architectures. HATEOAS links have the following attributes:

href	Complete url to make the call
method	Request method possible values are GET, POST or DELETE
rel	The link type
parameters	Map of request parameters name and value required for the call
valid_date	Date and time till when the url is valid

16. Supplementary FLOWS / Webhooks

Webhook call back for events for example when a customer's bank confirms a mandate setup, a customer stops an invoice for a recurring charge from the card issuer's side, or customer has revoked the mandate from the card issuer's side etc.

Call back to the merchant can occur in the following scenarios:

Mandate led notifications

- 1. Cancel Mandate Notification**
Customer cancels the mandate (from the card issuer's side).
- 2. Update Mandate Notification**
Status of the mandate received from the bank (bank account based mandate setup)
- 3. Mandate setup Notification**
Customer completes the setup e.g. UPI where the mandate setup process has been completed by customer on the UPI PSP app.

For the above use cases, BillDesk will POST the “mandate object” to your end-point. Merchant would need to then process the message and update its system based on latest status/ values received.

Invoice led notifications

1. Stop Invoice Notification

Customer puts a stop on the upcoming recurring charge (from the card issuer’s side).

BillDesk will POST the “invoice object” to your end-point. Merchant would need to then process the message and update its system based on the latest status/ values received.

Payment/charge transaction led notifications

1. Charge Update Notification

Status of the charge transaction received from the bank

BillDesk will POST the “transaction object” to your end-point. Merchant would need to then process the message and update its system based on the latest status/ values received.

Webhooks will be retried once after 30 minutes in case we receive connect/ read timeout or HTTP status other than ‘200’.

Similarly, there would be workflows that can emanate from the Merchant side with respect to:

1. Cancel Mandate
2. Update Mandate

Workflows for handling the Cancel Mandate and Update Mandate may vary basis the underlying payment instrument that was used to setup the mandate.

Event handling

Handling callback events correctly is crucial to making sure your integration’s business logic works as expected.

Acknowledge events immediately

If your callback script performs complex logic, or makes network calls, it’s possible that the script would time out before BillDesk sees its complete execution. Ideally, your callback handler code (acknowledging receipt of an event by returning a 2xx status code) is separate of any other logic you do for that event.

Handle duplicate events

Callback endpoints might occasionally receive the same event more than once. We advise you to guard against duplicated event receipts by making your event processing idempotent. One way of doing this is logging the events you’ve processed, and then not processing already-logged events.

Order of events

BillDesk does not guarantee delivery of events in the order in which they are generated. Your endpoint should not expect delivery of these events in any particular order and should handle it accordingly. You can also use the API to fetch any missing objects.